SOLUCIONARIO

EXAMEN PARCIAL

BMA 02 – SECCIÓN “P”

RESOLUCIÓN DEL PARCIAL

PROBLEMA 1A

I= $\int \frac{(1+\ln(x^2))}{\sqrt{x^4\ln(x^2)^2 - x^2\ln(x^4) - 35}}dx$ sea t= $x^2\ln(x)$; dt= $x(\ln(x^2)+1)$dx

I= $\int \frac{dt}{\sqrt{2x^4\ln(x)^2 - 4x^2\ln(x) - 35}}$; hacemos el cambio de variable

$\int \frac{dt}{\sqrt{2t^2-4t-35}}$;completamos cuadrados en el denominador

$* \ 2t^2 - 4t - 35$ = 2(t−1)^2-37 ; hacemos que (t-1)= $\frac{\sqrt{37}}{\sqrt{2}}$sec(α)

dt= $\frac{\sqrt{37}}{\sqrt{2}}$ sec(α) tan(α)dα ; y todo lo reemplazamos en I

$$I=\int \frac{\frac{\sqrt{37}}{\sqrt{2}} \sec(\alpha) \tan(\alpha) d\alpha}{\sqrt{37}\tan(\alpha)} = \frac{1}{\sqrt{2}}\int \sec(\alpha) d\alpha = \frac{1}{\sqrt{2}} \ln(\sec(\alpha)+) \tan(\alpha))+ C$$

Como resultado final y reemplazando $\alpha = \operatorname{arcsec}(\frac{\sqrt{2}}{\sqrt{37}}x^2\ln(x)-1)$

$$I=\frac{1}{\sqrt{2}} \ln(\sec(\operatorname{arcsec}(\frac{\sqrt{2}}{\sqrt{37}}x^2\ln(x)-1))+) \tan(\operatorname{arcsec}(\frac{\sqrt{2}}{\sqrt{37}}x^2\ln(x)-1)))+ C$$

PROBLEMA 1B.

Calcular la siguiente integral:

$$I = \int \frac{\cos(x).e^{\operatorname{sen}(x)}\,[\operatorname{sen}(x) + \ln(\cos(x))]\,[\cos^2(x) - \operatorname{sen}(x)]\tan(x)dx}{\operatorname{sen}(x)}$$

$$(\cos(x).e^{\operatorname{sen}(x)})' = e^{\operatorname{sen}(x)}\,[\cos^2(x) - \operatorname{sen}(x)]$$

$$\int [\operatorname{sen}(x) + \ln(\cos(x))]d\cos(x).e^{\operatorname{sen}(x)}$$

$$\mathrm{I} = \ln(\cos(x))]\cos(x).e^{\operatorname{sen}(x)} - \int [\cos(x).e^{\operatorname{sen}(x)}]d\,(\operatorname{sen}(x) + \ln(\cos(x)))$$

$$(\operatorname{sen}(x) + \ln(\cos(x)))' = \frac{\cos^2(x) - \operatorname{sen}(x)}{\cos(x)}$$

$$I=[\text{sen}(x) + \ln(\cos(x))]\cos(x).e^{\text{sen}(x)} - \int d\cos(x).e^{\text{sen}(x)}$$

RPTA

$$I=[\text{sen}(x) + \ln(\cos(x))]\cos(x).e^{\text{sen}(x)} - \cos(x).e^{\text{sen}(x)} + C$$

PROBLEMA 1B. (2da solución)

$$I = \int \frac{[cosx.e^{senx}][senx+\ln(cosx)][cos^2x-senx]tanx}{senx}dx$$

- $[cosx.e^{senx}]' = e^{senx}(cos^2x - senx)$

- $I = \int [senx + \ln(cosx)]d(cosx.e^{senx})$
- $I = [senx + \ln(cosx)]cosx.e^{senx}$- $\int cosx.e^{senx}d(senx + \ln(cosx)$

- $[senx + \ln(cosx)]' = \frac{cos^2x-senx}{cosx}$

RPTA

- $I = [senx + \ln(cosx)]cosx.e^{senx} - cosx.e^{senx} + cte$

PROBLEMA 1C.

$$\int \frac{e^{x}}{\sqrt{e^{x}-1}.\sqrt{2-e^{x}}[4\sqrt{e^{x}-1}+3\sqrt{2-e^{x}}]}dx$$

$$\textbf{Sabemos que}: d(e^{x}) = e^{x}dx$$

$$\textbf{Hacemos}: e^{x} = t$$

$$\int \frac{dt}{\sqrt{t-1}.\sqrt{2-t}[4\sqrt{t-1}+3\sqrt{2-t}]}$$

$$\textbf{Hacemos}: \sqrt{t-1} = senz \ y \ \sqrt{2-t} = cosz$$

$$\textbf{Sea}: sen^{2}z = t-1 \rightarrow 2.senz.cosz.dz = dt$$

$$\int \frac{2.senz.cosz.dz}{senz.cosz\ [4.senz+3.cosz]}$$

$$2\int \frac{dz}{[4.senz+3.cosz]}$$

$$\frac{2}{5}\int \frac{dz}{[\frac{4}{5}.senz+\frac{3}{5}.cosz]}$$

$$\boldsymbol{Notamos}: \left(\frac{4}{5}.senz + \frac{3}{5}.cosz\right) = sen(z + 37°)$$

$$\frac{2}{5}\int \frac{dz}{sen(z + 37°)}$$

$$\frac{2}{5}\int \csc(z + 37°).dz$$

$$\boldsymbol{Recordamos\ la\ integral\ fundamental}: \int cscx.dx = \ln|cscx - cotx| + C$$

$$\boldsymbol{Entonces}:$$

$$\frac{2}{5}\int \csc(z + 37°).d(z + 37°)$$

$$\frac{2}{5}\ln|\csc(z + 37°) - \cot(z + 37°)| + C$$

$$\boldsymbol{Recordamos\ que}: senz = \sqrt{t - 1} \ \rightarrow \ z = arcsen\left(\sqrt{t - 1}\right)$$

$$\boldsymbol{y}: \ t = e^{x} \rightarrow z = arcsen(\sqrt{e^{x} - 1})$$

RPTA

♣ $\frac{2}{5}\ln\left|\csc\left(arcsen(\sqrt{e^{x} - 1} + 37°\right) - \cot\left(arcsen(\sqrt{e^{x} - 1} + 37°\right)\right| + C$

PROBLEMA 2

$$I_p = \int x^p (m+nx)^{\frac{1}{2}}\, dx$$

$$u = x^p \qquad dv = (m+nx)^{\frac{1}{2}}\, dx$$

$$du = p x^{p-1} dx \qquad v = \frac{2}{3n}(m+nx)^{\frac{3}{2}}$$

$$I_p = \frac{2}{3n} x^p (m+nx)^{\frac{3}{2}} - \frac{2p}{3n}\int (m+nx)^{\frac{3}{2}} x^{p-1}\, dx$$

$$I_p = \frac{2}{3n} x^p (m+nx)^{\frac{3}{2}} - \frac{2p}{3n} m \int (m+nx)^{\frac{1}{2}} x^{p-1}\, dx - \frac{2p}{3n} n \int (m+nx)^{\frac{1}{2}} x^{p}\, dx$$

$$I_p = \frac{2}{3n} x^p (m+nx)^{\frac{3}{2}} - \frac{2p}{3n} m\, I_{p-1} - \frac{2p}{3} I_p$$

RPTA

$$I_p = \frac{2x^p (m+nx)^{\frac{3}{2}}}{n(3+2p)} - \frac{2pm}{n(3+2p)} I_{p-1}$$

PROBLEMA 3

Exprese el siguiente límite como una integral definida

$$\lim_{n\to\infty}\sum_{i=0}^{n}\left[\frac{48n^3+432n^2i+1296ni^2+1296i^3}{n^4}\right]\sin\left(\frac{4n^2+24ni+36i^2}{n^2}\right)$$

Solución:

Acomodamos la expresión:

$$\lim_{n\to\infty}\sum_{i=0}^{n}\left[48+432\left(\frac{i}{n}\right)+1296\left(\frac{i}{n}\right)^2+1296\left(\frac{i}{n}\right)^3\right]\cdot\sin\left(4+24\left(\frac{\mathrm{i}}{n}\right)+36\left(\frac{\mathrm{i}}{n}\right)^2\right)\cdot\frac{1}{n}$$

De donde:

$$a=0 \qquad\qquad x_i=a+i\,\mathrm{d}x$$

$$b=1$$

$$\mathrm{d}x=\frac{b-a}{n}=\frac{1-0}{n}=\frac{1}{n} \qquad\qquad x_i=0+\frac{i}{n}=\frac{i}{n}$$

Reemplazando:

$$\lim_{n\to\infty}\sum_{i=0}^{n}\left[\frac{48n^3+432n^2i+1296ni^2+1296i^3}{n^4}\right]\sin\left(\frac{4n^2+24ni+36i^2}{n^2}\right)$$

RPTA

$$\int_0^1 (48+432x+1296x^2+1296x^3)\sin(4+24x+36x^2)\cdot \mathrm{d}x$$

PROBLEMA 3 (2da solución)

Exprese el siguiente límite como una integral definida:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\frac{48n^3 + 432n^2 i + 1296ni^2 + 1296i^3}{n^4}\right] sen\left[\frac{4n^2 + 24ni + 36i^2}{n^2}\right]$$

<u>Solucion</u>:

Factorizamos "$\frac{6}{\mathrm{n}}$" *en la sumatoria.:*

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\frac{8n^3 + 72n^2 i + 216ni^2 + 216i^3}{n^3}\right] . \frac{6}{n} sen\left[\frac{4n^2 + 24ni + 36i^2}{n^2}\right]$$

Notamos que hay cuadrados y cubos perfectos, los reescribimos:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\frac{(2n + 6i)^3}{n^3}\right] sen\left[\frac{(2n + 6i)^2}{n^2}\right] . \frac{6}{n}$$

$Factorizamos\ los\ exponentes$:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\left(\frac{2n+6i}{n}\right)^3\right] sen\left[\left(\frac{2n+6i}{n}\right)^2\right] . \frac{6}{n}$$

$Simplificamos$:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\left(2+\frac{6i}{n}\right)^3\right] sen\left[\left(2+\frac{6i}{n}\right)^2\right] . \frac{6}{n}$$

$Sabemos\ que$:

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=0}^{n} f(x_i)\Delta x_i$$

$$Donde: \quad x_i = a + i.\Delta x_i$$

$$\Delta x_i = \frac{(b-a)}{n}$$

$Entonces$:

$$\Delta x_i = \frac{(b-a)}{n} = \frac{6}{n} \qquad \rightarrow \qquad b - a = 6$$

$Además$:

$$x_i = a + i.\Delta x_i = a + \frac{(b-a)i}{n} = 2 + \frac{6i}{n} \qquad \rightarrow \qquad a = 2 \quad y \quad b = 8$$

$Reemplazando$:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} [(x_i)^3] sen[(x_i)^2].\Delta x_i$$

$\underline{Por\ lo\ tanto, respuesta}$:

$$L = \lim_{n\to\infty} \sum_{i=0}^{n} \left[\frac{48n^3 + 432n^2 i + 1296ni^2 + 1296i^3}{n^4}\right] sen\left[\frac{4n^2 + 24ni + 36i^2}{n^2}\right] = \int_2^8 x^3 sen(x^2)dx$$

PROBLEMA 3 (3ra solución)

$$f(x) = \sum_{i=0}^{n}\left(\frac{48n^3 + 432n^2i + 1296ni^2 + 1296i^3}{n^4}\right) \sin[\frac{(4n^2 + 24ni + 36i^2)}{n^2}]$$

$$xi = a + \frac{(b-a)}{n}\,\text{i}$$

$$(48 + 432\frac{i}{n} + 1296\frac{i^2}{n^2} + 1296\frac{i^3}{n^3})\ sen\left(4 + 24\frac{i}{n} + 36\frac{i^2}{n^2}\right)\frac{1}{n}$$

$$\Delta\text{x} = \frac{(b-a)}{n}\ \text{i}$$

$$b = 1 \qquad a = 0$$

$$\left[\left(36\frac{i}{n} + 6\right)^2 + 12 + 1296\frac{i^3}{n^3}\right] sen\left[\left(6\frac{i}{n} + 2\right)^2\right]\frac{1}{n}$$

$$xi = \Delta\text{x} = \frac{i}{n}$$

RPTA

$$f(x) = \sum_{i=0}^{\infty} f(xi)\Delta\text{x} = \int_0^1 [(36x + 6)^2 + 12 + 1296x]\ sen[(6x + 2)^2]dx$$

PROBLEMA 4

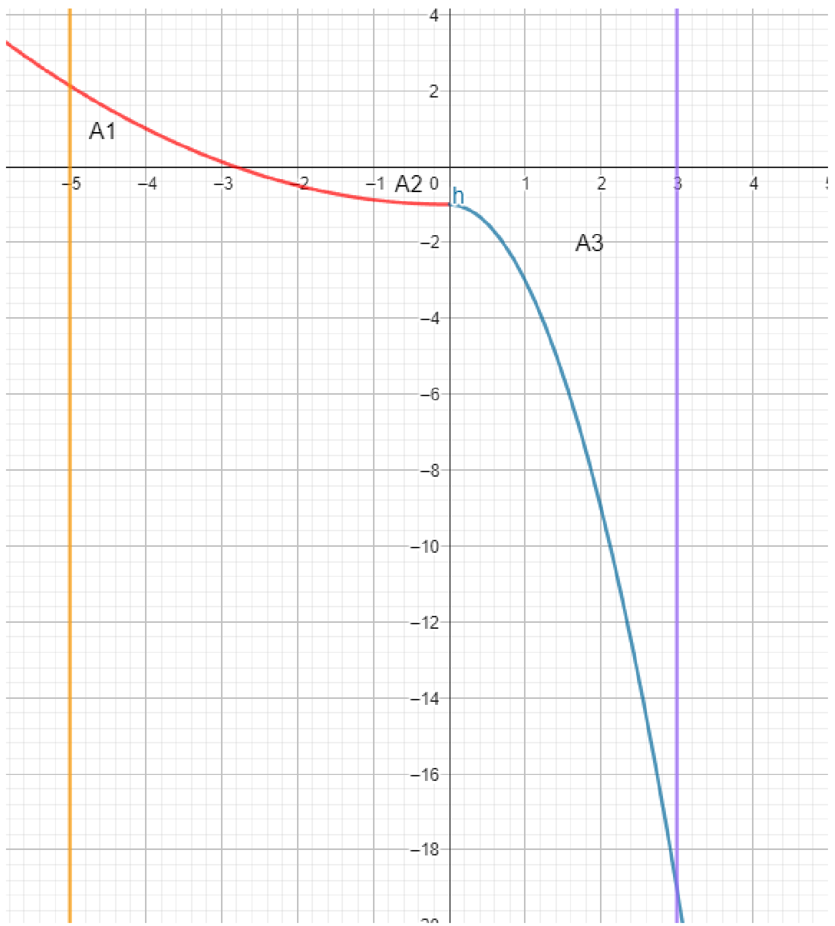
4. Empleando la definición de integral como límite de una suma, calculé el área de la región R limitada por la gráfica de la función f el eje X y las rectas $x = -5$, $x = 3$, siendo:

$$f(x) = \begin{cases} \frac{x^2}{8} - 1 \ , x < 0 \\ -2x^2 - 1 \ , \ x \geq 0 \end{cases}$$

(3,0 pts.)

Sea f(x) y su gráfica:

$$f(x) \begin{cases} \frac{(x^2)}{8} - 1; x<0 \\ -2x^2 - 1; x\geq 0 \end{cases}$$



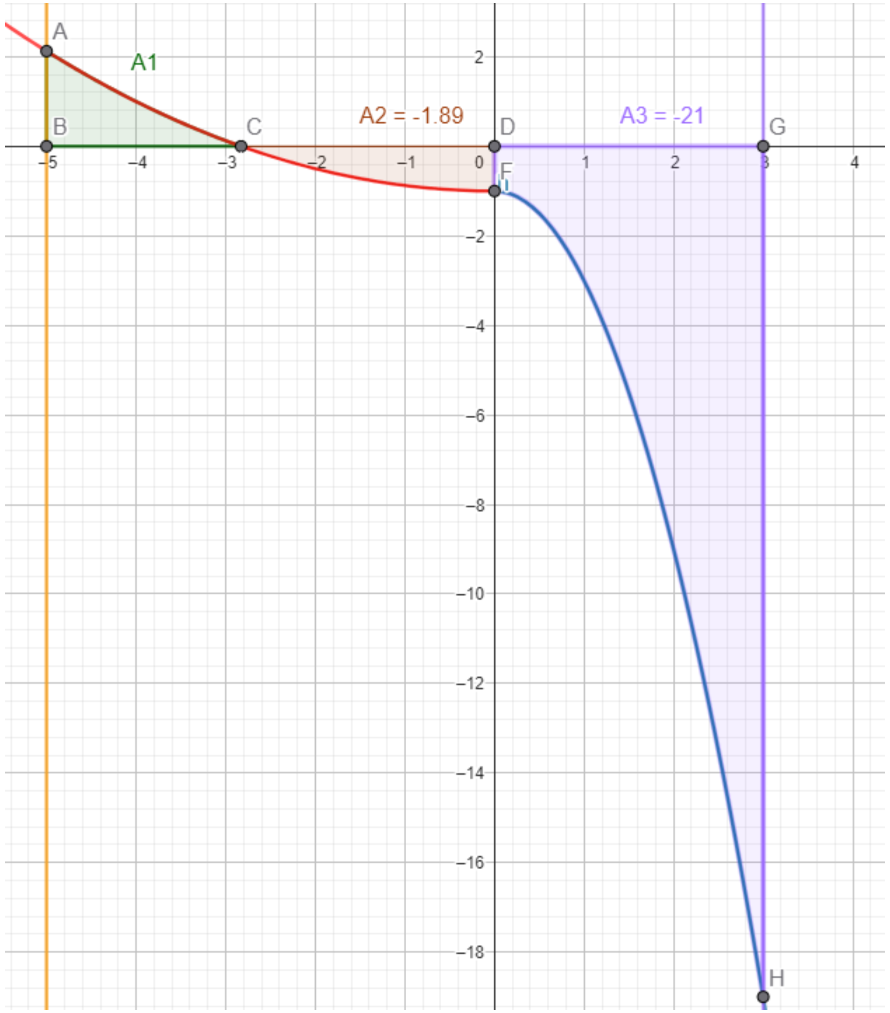
- $A_1 = \int_{-5}^{-2\sqrt{2}} (\frac{(x^2)}{8} - 1)\, dx = (\frac{x^3}{24} - x)\Big|_{-5}^{-2\sqrt{2}} = 2{,}09$

- $A_2 = -\int_{-5}^{-2\sqrt{2}} (\frac{(x^2)}{8} - 1)\, dx = (\frac{x^3}{24} - x)\Big|_{-2\sqrt{2}}^{0} = -1{,}89$

- $A_3 = -\int_0^3 (-2x^2 - 1)dx = (\frac{-2x^3}{3} - x)\Big|_0^3 = -21$

RPTA

$$AT = A_1 - A_2 - A_3$$
$$= 24,97$$

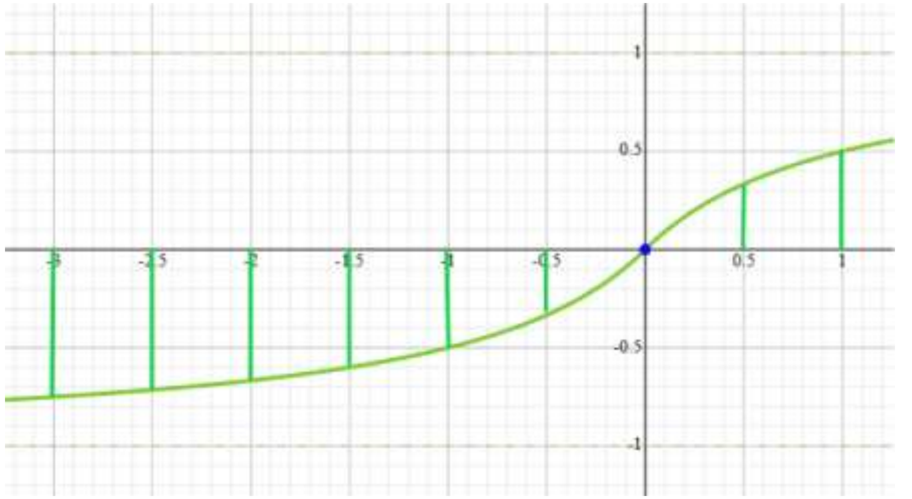


PROBLEMA 5

2.Hallar el valor aproximado de $\int_{-3}^{1} f(x)dx$ y una cota superior para su error

$$f(x) = \frac{x}{1+|x|} \qquad P = \left\{-3; -\frac{5}{2}; -2; -\frac{3}{2}; -1; -\frac{1}{2}; \frac{1}{2}; 1\right\}$$

I_i	Δx_i	M_i	m_i	$M_i \Delta x_i$	$m_i \Delta x_i$
$\left[-3; -\frac{5}{2}\right]$	$1/2$	$f\left(-\frac{5}{2}\right) = -5/7$	$f(-3) = -3/4$	$-5/14$	$-3/8$
$\left[-\frac{5}{2}; -2\right]$	$1/2$	$f(-2) = -2/3$	$f\left(-\frac{5}{2}\right) = -5/7$	$-1/3$	$-5/14$
$\left[-2; -\frac{3}{2}\right]$	$1/2$	$f\left(-\frac{3}{2}\right) = -3/5$	$f(-2) = -2/3$	$-3/10$	$-1/3$
$\left[-\frac{3}{2}; -1\right]$	$1/2$	$f(-1) = -1/2$	$f\left(-\frac{3}{2}\right) = -3/5$	$-1/4$	$-3/10$
$\left[-1; -\frac{1}{2}\right]$	$1/2$	$f\left(-\frac{1}{2}\right) = -1/3$	$f(-1) = -1/2$	$-1/6$	$-1/4$
$\left[-\frac{1}{2}; \frac{1}{2}\right]$	1	$f\left(\frac{1}{2}\right) = 1/3$	$f\left(-\frac{1}{2}\right) = -1/3$	$1/3$	$-1/6$
$\left[\frac{1}{2}; 1\right]$	$1/2$	$f(1) = 1/2$	$f\left(\frac{1}{2}\right) = 1/3$	$1/4$	$1/3$



$$U(f,p)=\sum_{i=1}^{n} M_i \Delta x_i \rightarrow U(f,p)=-\frac{5}{14}-\frac{1}{3}-\frac{3}{10}-\frac{1}{4}-\frac{1}{6}+\frac{1}{3}+\frac{1}{4}=-\frac{173}{210}$$

$$L(f,p)=\sum_{i=1}^{n} m_i \Delta x_i \rightarrow L(f,p)=-\frac{3}{8}-\frac{5}{14}-\frac{1}{3}-\frac{3}{10}-\frac{1}{4}-\frac{1}{6}+\frac{1}{3}=-\frac{1217}{840}$$

$$\int_{-3}^{1} f(x)dx \approx \left|\frac{U(f,p)+L(f,p)}{2}\right|$$

$$\approx \left|\frac{-\frac{173}{210}-\frac{1217}{840}}{2}\right| \approx |-1{,}136309524| \approx 1{,}136309524$$

Ahora la cota superior para su error (ε)

$$\varepsilon \leq \frac{U(f,p)-L(f,p)}{2}$$

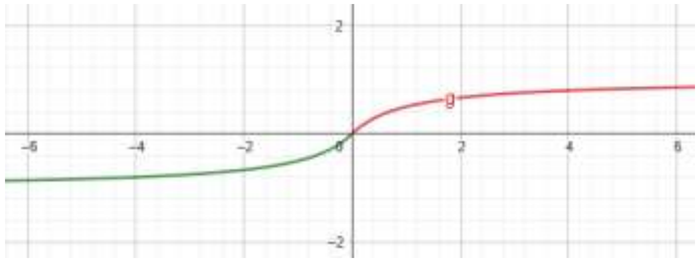
$$\varepsilon \leq \frac{-\frac{173}{210}+\frac{1217}{840}}{2}$$

$$\varepsilon \leq 0.3125$$

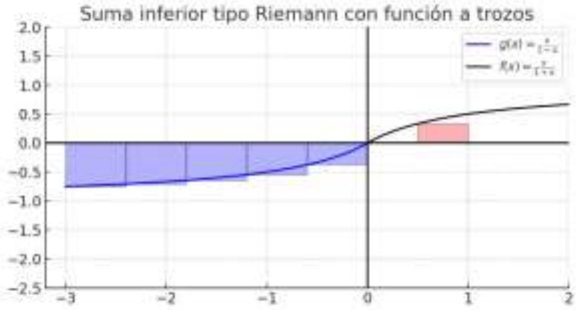
PROBLEMA (2da solución)

5. Aproxima el valor de la integral $\int_{-3}^{1} \frac{x}{1+|x|}dx$ y dar una cota de su error, aplicando la suma superior e inferior mediante la partición P=$\{-3; -\frac{5}{2}; -2; -\frac{3}{2}; -1; -\frac{1}{2}; \frac{1}{2}; 1\}$

Idea gráfica:



Para L:



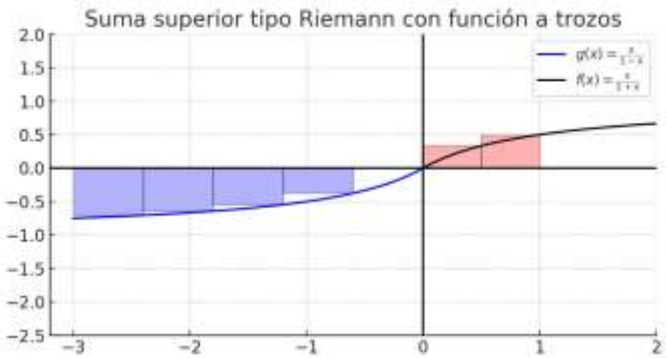
	-3	$-\frac{5}{2}$	-2	$-\frac{3}{2}$	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1
$f(\ )$	$-\frac{3}{4}$	$-\frac{5}{7}$	$-\frac{2}{3}$	$-\frac{3}{5}$	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{2}$

$$L_1 = \frac{1}{2}\left[f\left(-\frac{5}{2}\right) + f(-2) + f\left(-\frac{3}{2}\right) + f(-1) + f\left(-\frac{1}{2}\right)\right] \Rightarrow L_1 = \frac{-197}{140}$$

$$L_2 = \frac{1}{2}\left[f\left(\frac{1}{2}\right)\right] \Rightarrow L_2 = \frac{1}{6}$$

$$L = -L_1 + L_2 \quad \Rightarrow \quad L = -(\frac{-197}{140}) + \frac{1}{6} \quad \Rightarrow \quad L = \frac{661}{420}$$

Para u:



$$U_1 = \frac{1}{2}[f(-3) + f(-\frac{5}{2}) + f(-\frac{3}{2}) + f(-2) + f(-\frac{3}{2}) + f(-\frac{1}{2})] \quad \Rightarrow \quad U_1 = \frac{-499}{280}$$

$$U_2 = \frac{1}{2}[f(\frac{1}{2}) + f(1)] \quad \Rightarrow \quad U_2 = \frac{5}{12}$$

$$U = -U_1 + U_2 \quad \Rightarrow \quad U = -(\frac{-499}{5}$$

1199

$$\overline{280}$$

$$I = \int_{-3}^{1} \frac{d}{1 + |\ |} \cong \frac{L + U}{2} \quad \Rightarrow \quad 280 \; I = \quad) + \quad \Rightarrow \quad \Rightarrow \quad I = 2,93$$

Con un error:

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$$error \leq \frac{L + U}{2} = \frac{65}{24}$$

4
9
1
9
1
6
8
0